

Table 2.3 Characteristics of Light and Transmission Electron Microscopes

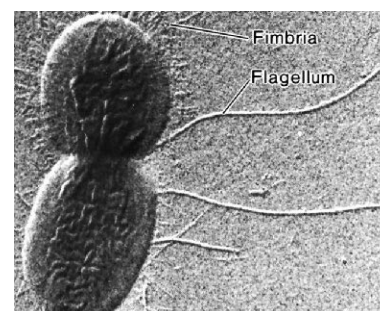
Feature	Light Microscope	Electron Microscope
Highest practical magnification	About 1,000–1,500	Over 100,000
Best resolution ^a	0.2 μm	0.5 nm
Radiation source	Visible light	Electron beam
Medium of travel	Air	High vacuum
Type of lens	Glass	Electromagnet
Source of contrast	Differential light absorption	Scattering of electrons
Focusing mechanism	Adjust lens position mechanically	Adjust current to the magnetic lens
Method of changing magnification	Switch the objective lens or eyepiece	Adjust current to the magnetic lens
Specimen mount	Glass slide	Metal grid (usually copper)

^aThe resolution limit of a human eye is about 0.2 mm.

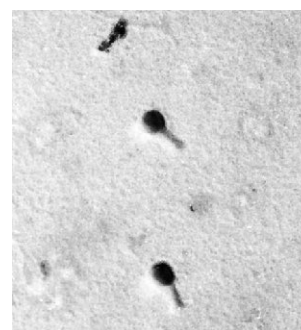
thin sections with solutions of heavy metal salts like lead citrate and uranyl acetate. The lead and uranium ions bind to cell structures and make them more electron opaque, thus increasing contrast in the material. Heavy osmium atoms from the osmium tetroxide fixative also “stain” cells and increase their contrast. The stained thin sections are then mounted on tiny copper grids and viewed.

Although specimens normally are embedded in plastic and thin sectioned to reveal the internal structure of the smallest cell, there are other ways in which microorganisms and smaller objects can be readied for viewing. One very useful technique is negative staining. The specimen is spread out in a thin film with either phosphotungstic acid or uranyl acetate. Just as in negative staining for light microscopy, heavy metals do not penetrate the specimen but render the background dark, whereas the specimen appears bright in photographs. Negative staining is an excellent way to study the structure of viruses, bacterial gas vacuoles, and other similar material. A microorganism also can be viewed after **shadowing** with metal. It is coated with a thin film of platinum or other heavy metal by evaporation at an angle of about 45° from horizontal so that the metal strikes the microorganism on only one side. The area coated with metal scatters electrons and appears light in photographs, whereas the uncoated side and the shadow region created by the object is dark (**figure 2.24**). The specimen looks much as it would if light were shining on it to cast a shadow. This technique is particularly useful in studying virus morphology, bacterial flagella, and plasmids (*see chapter 13*).

The TEM will also disclose the shape of organelles within microorganisms if specimens are prepared by the **freeze-etching** procedure. Cells are rapidly frozen in liquid nitrogen and then warmed to -100°C in a vacuum chamber. Next a knife that has been pre-cooled with liquid nitrogen (-196°C) fractures the frozen cells, which are very brittle and break along lines of greatest weakness, usually down the middle of internal membranes (**figure 2.25**). The specimen is left in the high vacuum for a minute or more so that some of the ice can sublime away and uncover more structural detail (sometimes this etching step is eliminated). Finally, the exposed surfaces are shadowed and coated with layers of platinum



(a)



(b)

Figure 2.24 Specimen Shadowing for the TEM. Examples of specimens viewed in the TEM after shadowing with uranium metal. (a) *Proteus mirabilis* ($\times 42,750$); note flagella and fimbriae. (b) T4 coliphage ($\times 72,000$).

and carbon to form a replica of the surface. After the specimen has been removed chemically, this replica is studied in the TEM and provides a detailed, three-dimensional view of intracellular structure (**figure 2.26**). An advantage of freeze-etching is that it minimizes the danger of artifacts because the cells are frozen quickly rather than being subjected to chemical fixation, dehydration, and plastic embedding.

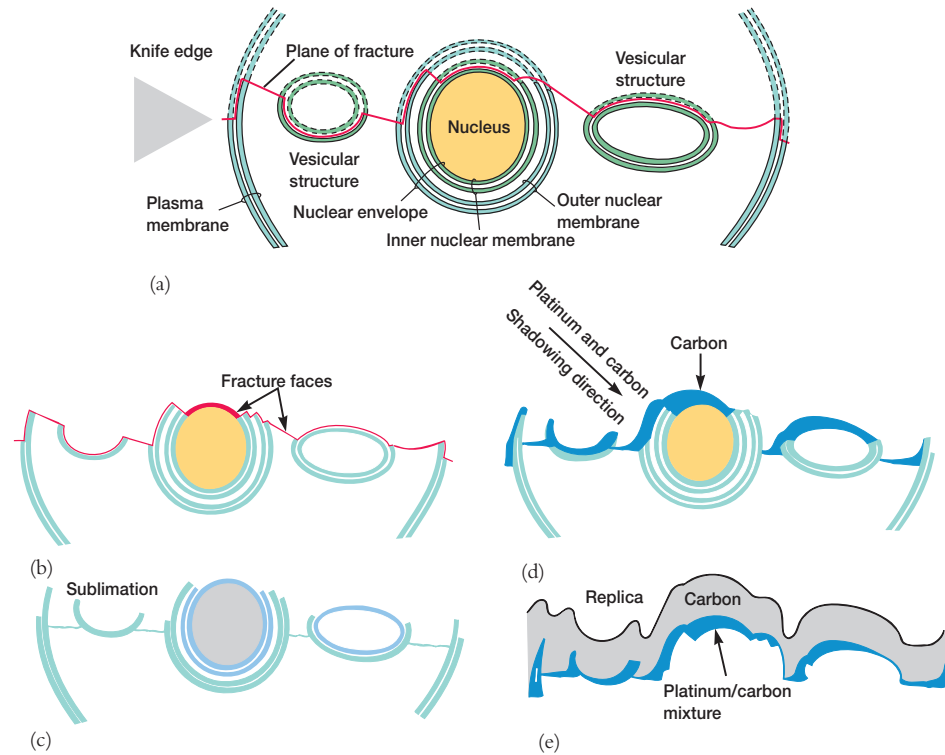


Figure 2.25 The Freeze-Etching Technique. In steps (a) and (b), a frozen eucaryotic cell is fractured with a cold knife. Etching by sublimation is depicted in (c). Shadowing with platinum plus carbon and replica formation are shown in (d) and (e). See text for details.

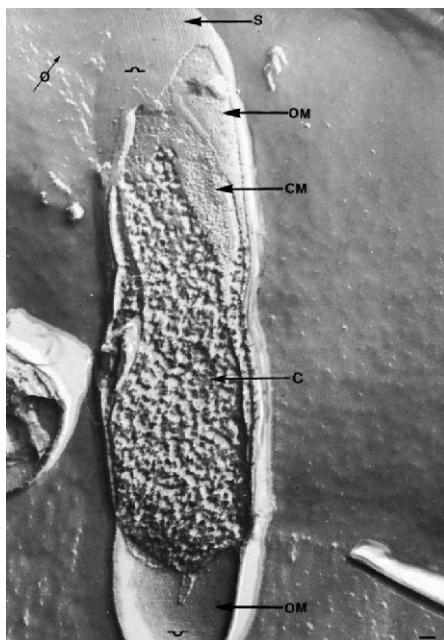


Figure 2.26 Example of Freeze-Etching. A freeze-etched preparation of the bacterium *Thiobacillus kabobis*. Note the differences in structure between the outer surface, S; the outer membrane of the cell wall, OM; the cytoplasmic membrane, CM; and the cytoplasm, C. Bar = 0.1 μm .

The Scanning Electron Microscope

The previously described microscopes form an image from radiation that has passed through a specimen. More recently the **scanning electron microscope (SEM)** has been used to examine the surfaces of microorganisms in great detail; many instruments have a resolution of 7 nm or less. The SEM differs from other electron microscopes in producing an image from electrons emitted by an object's surface rather than from transmitted electrons.

Specimen preparation is easy, and in some cases air-dried material can be examined directly. Most often, however, microorganisms must first be fixed, dehydrated, and dried to preserve surface structure and prevent collapse of the cells when they are exposed to the SEM's high vacuum. Before viewing, dried samples are mounted and coated with a thin layer of metal to prevent the buildup of an electrical charge on the surface and to give a better image.

The SEM scans a narrow, tapered electron beam back and forth over the specimen (**figure 2.27**). When the beam strikes a particular area, surface atoms discharge a tiny shower of electrons called secondary electrons, and these are trapped by a special detector. Secondary electrons entering the detector strike a scintillator causing it to emit light flashes that a photomultiplier converts to an electrical current and amplifies. The signal is sent to a cathode-ray tube and produces an image like a television picture, which can be viewed or photographed.

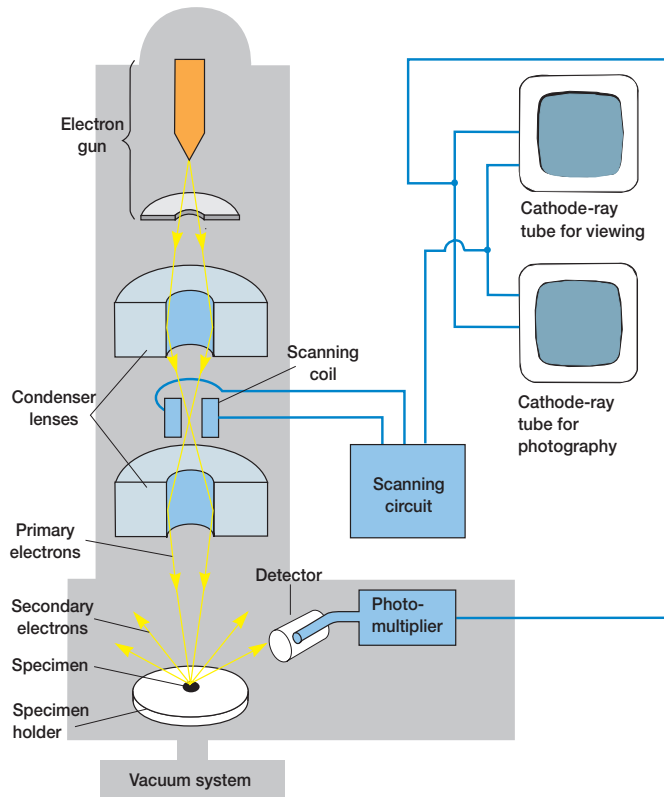


Figure 2.27 The Scanning Electron Microscope.

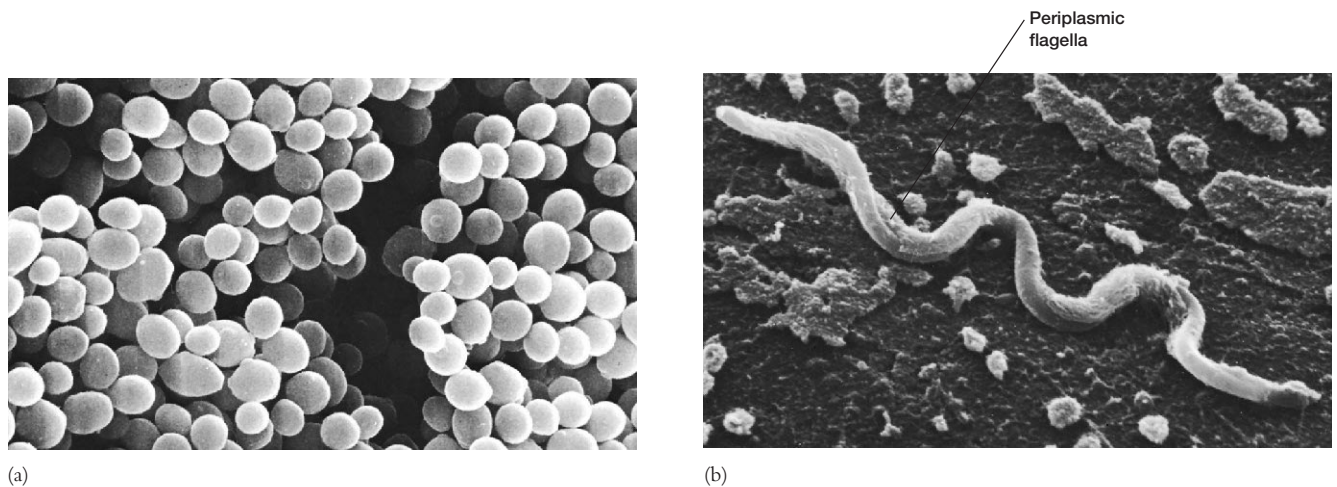


Figure 2.28 Scanning Electron Micrographs of Bacteria. (a) *Staphylococcus aureus* ($\times 32,000$). (b) *Cristispira*, a spirochete from the crystalline style of the oyster, *Ostrea virginica*. The axial fibrils or periplasmic flagella are visible around the protoplasmic cylinder ($\times 6,000$).

The number of secondary electrons reaching the detector depends on the nature of the specimen's surface. When the electron beam strikes a raised area, a large number of secondary electrons enter the detector; in contrast, fewer electrons escape a depression in the surface and reach the detector. Thus raised

areas appear lighter on the screen and depressions are darker. A realistic three-dimensional image of the microorganism's surface with great depth of focus results (**figure 2.28**). The actual in situ location of microorganisms in ecological niches such as the human skin and the lining of the gut also can be examined.

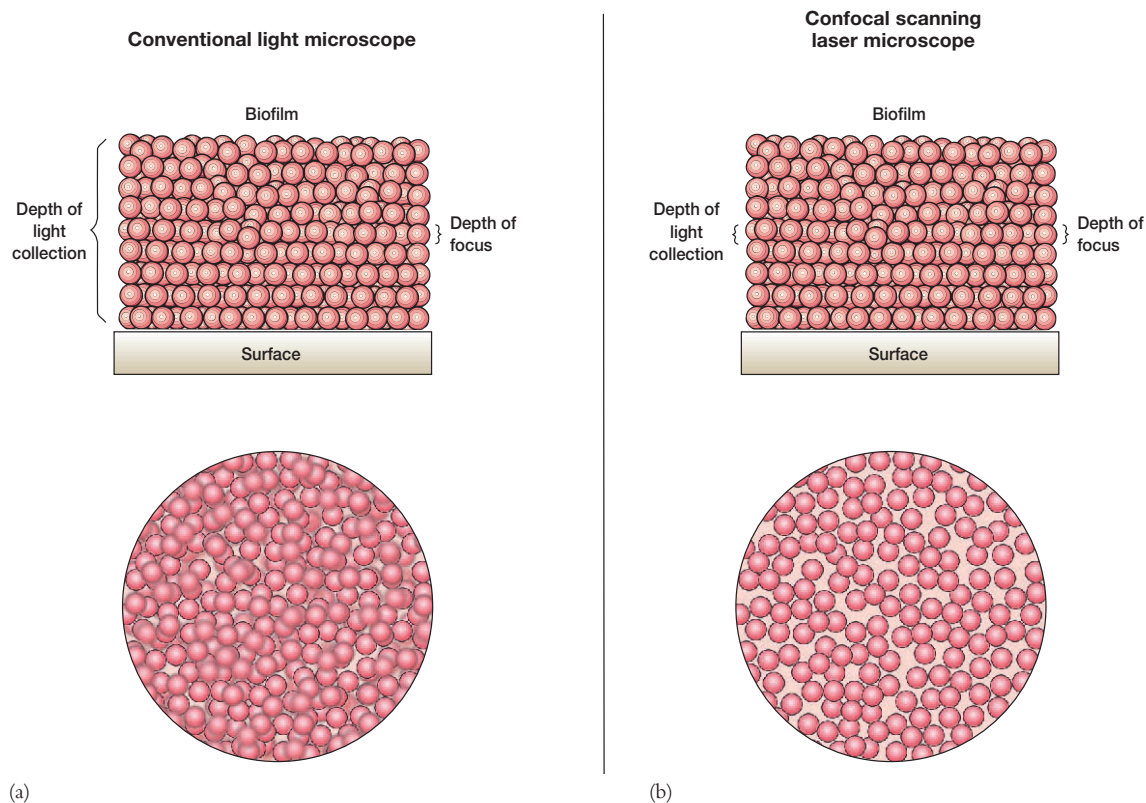


Figure 2.29 Confocal Scanning Laser Microscopy: Light Collection Depth and Image Clarity. (a) Conventional light microscopic observation. (b) Confocal scanning laser microscopic observation.

1. Why does the transmission electron microscope have much greater resolution than the light microscope? Describe in general terms how the TEM functions.
2. Describe how specimens are prepared for viewing in the TEM. How are sections usually stained to increase contrast? What are negative staining, shadowing, and freeze-etching?
3. How does the scanning electron microscope operate and in what way does its function differ from that of the TEM? The SEM is used to study which aspects of morphology?

2.5 Newer Techniques in Microscopy

Confocal Microscopy

A conventional light microscope, which uses a mixed wavelength light source and illuminates a large area of the specimen, will have a relatively great depth of field. Even if not in focus, images of bacteria from all levels within the field will be visible. These will include cells above, in, and below the plane of focus (figure 2.29). As a result the image can be murky, fuzzy, and crowded.

The solution to this problem is the **confocal scanning laser microscope (CSLM)** or confocal microscope. Fluorescently stained specimens are usually examined. A focused laser beam strikes a point in the specimen (figure 2.30). Light from the illuminated spot is focused by an objective lens onto a plane above the objective. An aperture above the objective lens blocks out stray light from parts of the specimen that lie above and below the plane of focus. The laser is scanned over a plane in the specimen (beam scanning) or the stage is moved (stage scanning) and a detector measures the illumination from each point to produce an image of the optical section. When many optical sections are scanned, a computer can combine them to form a three-dimensional image from the digitized signals. This image can be measured and analyzed quantitatively.

The confocal microscope improves images in two ways. First, illumination of one spot at a time reduces interference from light scattering by the rest of the specimen. Second, the aperture above the objective lens blocks out stray light as previously mentioned. Consequently the image has excellent contrast and resolution. A depth of 1 μm or less in a thick preparation can be directly observed. Special computer software is used to create high-resolution, three-dimensional images of cell structures and complex specimens such as biofilms (figure 2.31).

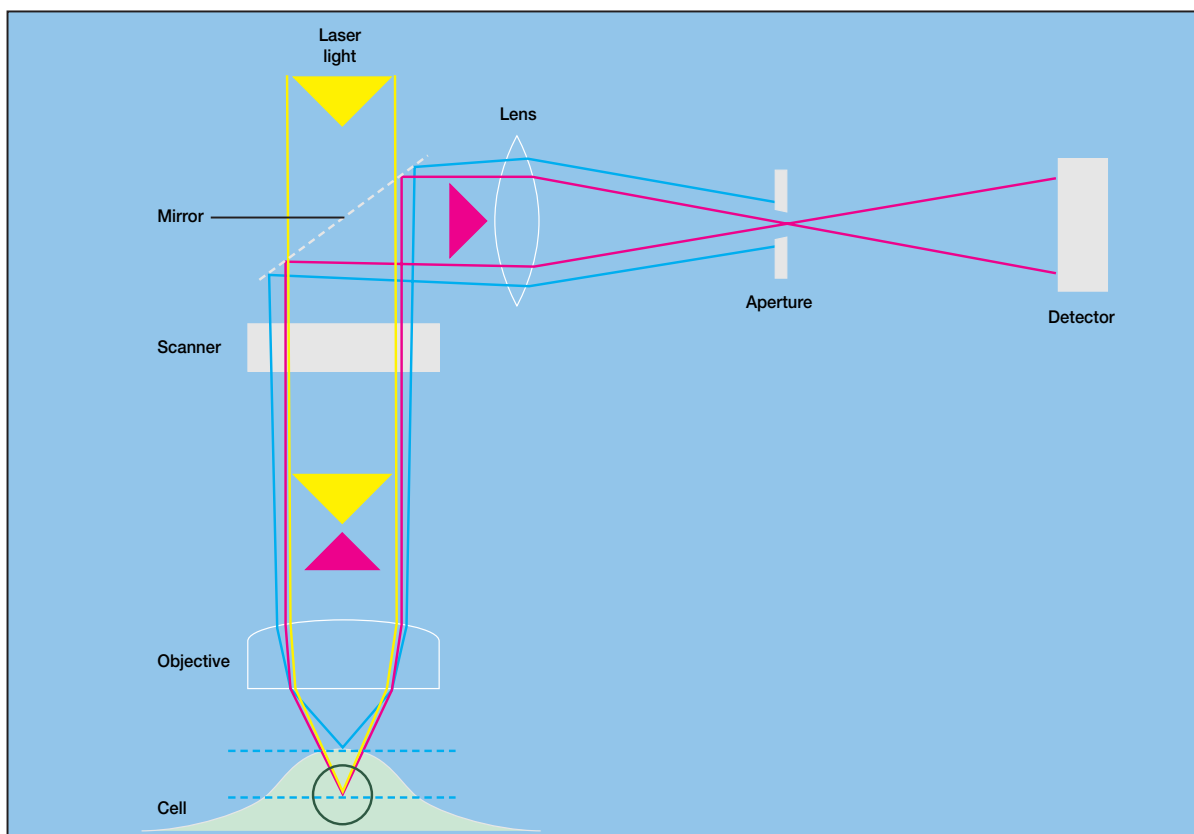
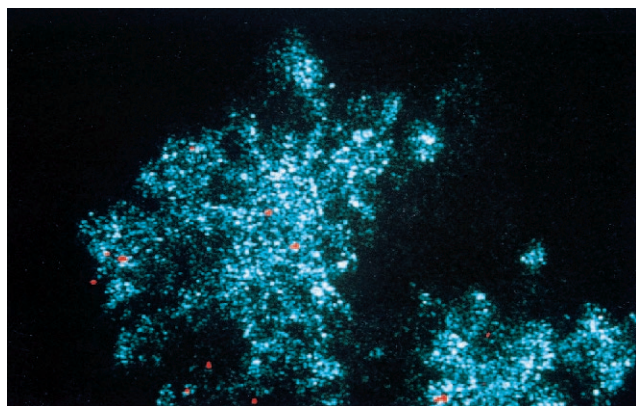
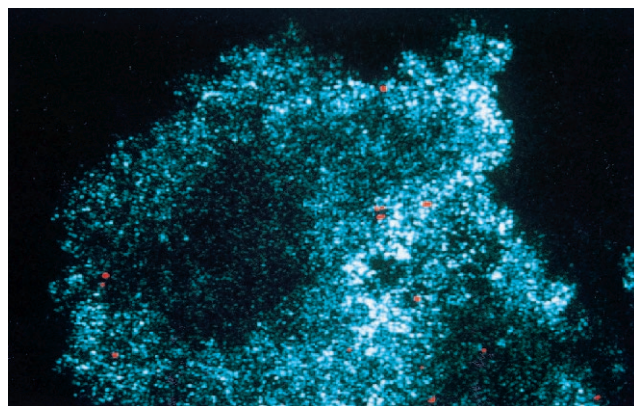


Figure 2.30 A Ray Diagram of a Confocal Laser Scanning Microscope. The yellow lines represent laser light used for illumination. Red lines symbolize the light arising from the plane of focus, and the blue lines stand for light from parts of the specimen above and below the focal plane. See text for explanation.



(a)



(b)

Figure 2.31 Confocal Images at Various Depths below the Top of a Biofilm. (a) 20 μm . (b) 40 μm . Each of these confocal images—which combines fluorescent and reflection images—has a depth of 2 μm and shows red-colored fluorescent tracer beads.

Scanning Probe Microscopy

Although light and electron microscopes have become quite sophisticated and reached an advanced state of development, powerful new microscopes are still being created. A new class of microscopes, called **scanning probe microscopes**, measure surface features by moving a sharp probe over the object's surface. The **scanning tunneling microscope**, invented in 1980, is an excellent example of a scanning probe microscope. It can achieve magnifications of 100 million and allow scientists to view atoms on the surface of a solid. The electrons surrounding surface atoms tunnel or project out from the surface boundary a very short distance. The scanning tunneling microscope has a needlelike probe with a point so sharp that often there is only one atom at its tip. The probe is lowered toward the specimen surface until its electron cloud just touches that of the surface atoms. If a small voltage is applied between the tip and specimen, electrons flow through a narrow channel in the electron clouds. This tunneling current, as it is called, is extraordinarily sensitive to distance and will decrease about a thousandfold if the probe is moved away from the surface by a distance equivalent to the diameter of an atom.

The arrangement of atoms on the specimen surface is determined by moving the probe tip back and forth over the surface while keeping it at a constant height by adjusting the probe distance to maintain a steady tunneling current. As the tip moves up and down while following the surface contours, its motion is recorded and analyzed by a computer to create an accurate three-dimensional image of the surface atoms. The surface map can be displayed on a computer screen or plotted on paper. The resolution is so great that individual atoms are observed easily. The microscope's inventors, Gerd Binnig and Heinrich Rohrer, shared the 1986 Nobel Prize in Physics for their work, together with Ernst Ruska, the designer of the first transmission electron microscope.

The scanning tunneling microscope will likely have a major impact in biology. Recently it has been used to directly view DNA (**figure 2.32**). Since the microscope can examine objects when they are immersed in water, it may be particularly useful in studying biological molecules.

More recently a second type of scanning probe microscope has been developed. The **atomic force microscope** moves a sharp probe over the specimen surface while keeping the distance between the probe tip and the surface constant. It does this by exerting a very small amount of force on the tip, just enough to maintain a constant distance but not enough force to damage the surface. The vertical motion of the tip usually is followed by measuring the deflection of

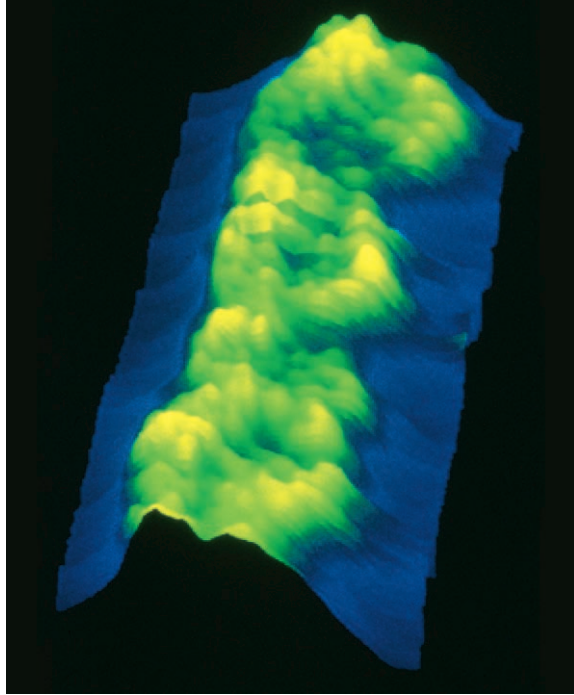


Figure 2.32 Scanning Tunneling Microscopy of DNA. The DNA double helix with approximately three turns shown (false color; $\times 2,000,000$).

a laser beam that strikes the lever holding the probe. Unlike the scanning tunneling microscope, the atomic force microscope can be used to study surfaces that do not conduct electricity well. The atomic force microscope has been used to study the interactions between the *E. coli* GroES and GroEL chaperonin proteins, to map plasmids by locating restriction enzymes bound to specific sites, and to follow the behavior of living bacteria and other cells.

1. How does a confocal microscope operate and why does it provide better images of thick specimens than the standard compound microscope?
2. Briefly describe the scanning probe microscope and its most popular versions, the scanning tunneling microscope and the atomic force microscope. What are these microscopes used for?

Summary

1. A light ray moving from air to glass, or vice versa, is bent in a process known as refraction. Lenses focus light rays at a focal point and magnify images (**figure 2.2**).
2. In a compound microscope like the bright-field microscope, the primary image is formed by an objective lens and enlarged by the eyepiece or ocular lens to yield a virtual image (**figure 2.3**).
3. A substage condenser focuses a cone of light on the specimen.
4. Microscope resolution increases as the wavelength of radiation used to illuminate the specimen decreases. The maximum resolution of a light microscope is about $0.2\ \mu\text{m}$.
5. The dark-field microscope uses only refracted light to form an image (**figure 2.7**), and objects glow against a black background.
6. The phase-contrast microscope converts variations in the refractive index and density of cells into changes in light intensity and thus makes colorless, unstained cells visible (**figure 2.9**).
7. The differential interference contrast microscope uses two beams of light to create high-contrast, three-dimensional images of live specimens.

8. The fluorescence microscope illuminates a fluorochrome-labeled specimen and forms an image from its fluorescence (**figure 2.12**).
9. Specimens usually must be fixed and stained before viewing them in the bright-field microscope.
10. Most dyes are either positively charged basic dyes or negative acid dyes and bind to ionized parts of cells.
11. In simple staining a single dye mixture is used to stain microorganisms.
12. Differential staining procedures like the Gram stain and acid-fast stain distinguish between microbial groups by staining them differently.
13. Some staining techniques are specific for particular structures like bacterial capsules, flagella, and endospores.
14. The transmission electron microscope uses magnetic lenses to form an image from electrons that have passed through a very thin section of a specimen (**figure 2.23**). Resolution is high because the wavelength of electrons is very short.
15. Thin section contrast can be increased by treatment with solutions of heavy metals like osmium tetroxide, uranium, and lead.
16. Specimens are also prepared for the TEM by negative staining, shadowing with metal, or freeze-etching.
17. The scanning electron microscope (**figure 2.27**) is used to study external surface features of microorganisms.
18. The confocal scanning laser microscope (**figure 2.29**) is used to study thick, complex specimens. Scanning probe microscopes can visualize molecules and cells.

Key Terms

- | | | |
|--|----------------------------|---|
| acid dyes 27 | flagella staining 30 | phase-contrast microscope 22 |
| acid-fast staining 28 | fluorescence microscope 25 | refraction 18 |
| atomic force microscope 38 | fluorescent light 25 | refractive index 18 |
| basic dyes 27 | fluorochromes 25 | resolution 20 |
| bright-field microscope 19 | focal length 18 | scanning electron microscope (SEM) 34 |
| chromophore groups 27 | focal point 18 | scanning probe microscope 38 |
| confocal scanning laser microscope (CSLM) 36 | freeze-etching 33 | scanning tunneling microscope 38 |
| dark-field microscopy 22 | Gram stain 28 | shadowing 33 |
| dark-phase-contrast microscopy 24 | mordant 28 | simple staining 28 |
| differential interference contrast (DIC) microscope 25 | negative staining 28 | spore staining 30 |
| differential staining procedures 28 | numerical aperture 20 | substage condenser 19 |
| eyepieces 19 | objectives 19 | transmission electron microscope (TEM) 30 |
| fixation 27 | oculars 19 | working distance 21 |
| | parfocal 20 | |

Questions for Thought and Review

1. How are real and virtual images produced in a light microscope? Which one is a person actually seeing?
2. If a specimen is viewed using a 43 $\times$ objective in a microscope with a 15 $\times$ eyepiece, how many times has the image been magnified?
3. Why don't most light microscopes use 30 $\times$ eyepieces for greater magnification?
4. Describe the two general types of fixation. Which would you normally use for bacteria? For protozoa?
5. Why would one expect basic dyes to be more effective under alkaline conditions?
6. What step in the Gram-stain procedure could be dropped without losing the ability to distinguish between gram-positive and gram-negative bacteria? Why?
7. Why must the TEM use a high vacuum and very thin sections?
8. Material is often embedded in paraffin before sectioning for light microscopy. Why can't this approach be used when preparing a specimen for the TEM?
9. Under what circumstances would it be desirable to prepare specimens for the TEM by use of negative staining? Shadowing? Freeze-etching?
10. Compare the microscopes described in this chapter—bright-field, dark-field, phase-contrast, DIC, fluorescence, TEM, SEM, confocal, and scanning probe—in terms of the images they provide and the purposes for which they are most often used.
11. Describe briefly how the scanning probe microscope operates. For what is it used? Distinguish between the two types of scanning probe microscopes with respect to their mechanism of operation.
12. Prepare a summary table showing the advantages of each type of microscope described in the chapter.

Critical Thinking Questions

1. If you prepared a sample of a specimen for light microscopy, stained with the Gram stain, and failed to see anything when you looked through your light microscope, list the things that you may have done incorrectly.
2. In a journal article, find an example of a light micrograph, a scanning or transmission electron micrograph, or a confocal image. Discuss why the figure was included in the article and why that particular type of microscopy was the method of choice for the research. What other figures would you like to see used in this study? Outline the steps that the investigators would take in order to obtain such photographs or figures.

Additional Reading

General

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2.4 Electron Microscopy

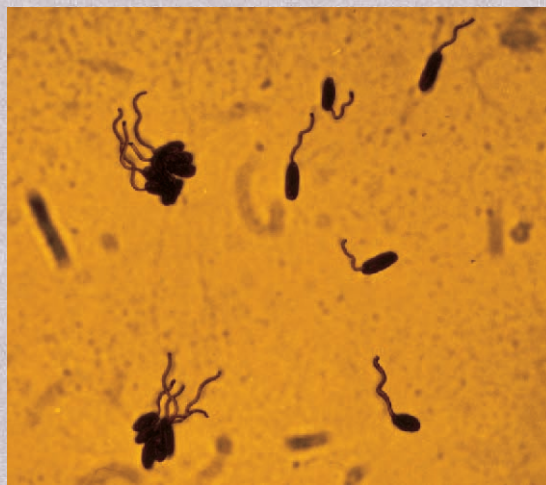
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CHAPTER 3

Prokaryotic Cell Structure and Function



Bacterial species may differ in their patterns of flagella distribution. These *Pseudomonas* cells have a single polar flagellum used for locomotion.

Outline

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- 3.8 The Bacterial Endospore 68

Concepts

1. Bacteria are small and simple in structure when compared with eucaryotes, yet they often have characteristic shapes and sizes.
2. Although they have a plasma membrane, which is required by all living cells, bacteria generally lack extensive, complex, internal membrane systems.
3. The cytoplasmic matrix typically contains several constituents that are not membrane-enclosed: inclusion bodies, ribosomes, and the nucleoid with its genetic material.
4. The prokaryotic cell wall almost always has peptidoglycan and is chemically and morphologically complex. Most bacteria can be divided into gram-positive and gram-negative groups based on their cell wall structure and response to the Gram stain.
5. Components like capsules and fimbriae are located outside the cell wall. One of these is the flagellum, which many bacteria use like a propeller to swim toward attractants and away from repellents.
6. Some bacteria form resistant endospores to survive harsh environmental conditions in a dormant state.

The era in which workers tended to look at bacteria as very small bags of enzymes has long passed.

—Howard J. Rogers

Even a superficial examination of the microbial world shows that bacteria are one of the most important groups by any criterion: numbers of organisms, general ecological importance, or practical importance for humans. Indeed, much of our understanding of phenomena in biochemistry and molecular biology comes from research on bacteria. Although considerable space is devoted to eucaryotic microorganisms, the major focus is on procaryotes. Therefore the unit on microbial morphology begins with the structure of procaryotes. As mentioned in chapter 1 (*see p. 11*), there are two quite different groups of procaryotes: Bacteria and Archaea. This chapter focuses primarily on bacterial morphology; chapter 20 will discuss archaeal cell structure and composition. A comment about nomenclature is necessary to avoid confusion. The word procaryote will be used in a general sense to include both the bacteria and archaea; the term bacterium will refer specifically to bacteria. [Eucaryotes, procaryotes, and the composition of the microbial world \(pp. 11; 91–92\). The Archaea \(pp. 450–65\)](#)

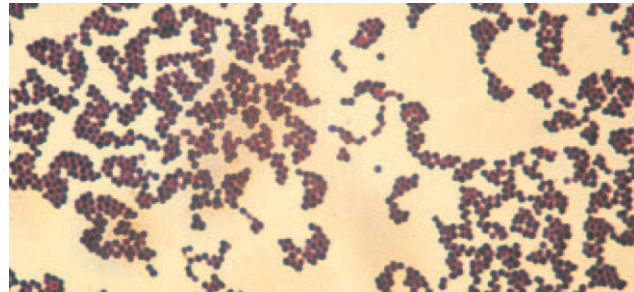
3.1 An Overview of Prokaryotic Cell Structure

Because much of this chapter is devoted to a discussion of individual cell components, a preliminary overview of the procaryotic cell as a whole is in order.

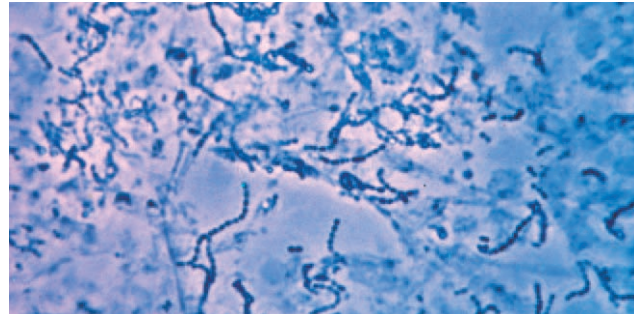
Size, Shape, and Arrangement

One might expect that small, relatively simple organisms like procaryotes would be uniform in shape and size. Although it is true that many procaryotes are similar in morphology, there is a remarkable amount of variation due to differences in genetics and ecology. (**figures 3.1 and 3.2**; *see also figures 2.8 and 2.15*). Major morphological patterns are described here, and interesting variants are mentioned in the procaryotic survey (*see chapters 20–24*).

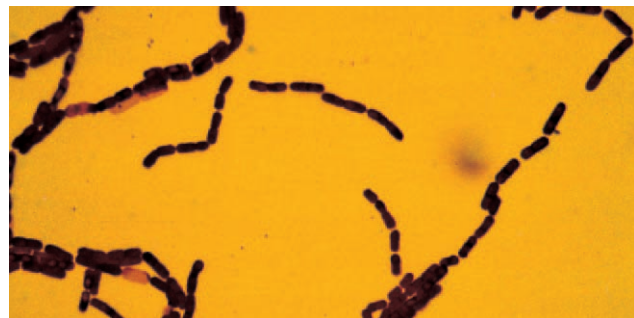
Most commonly encountered bacteria have one of two shapes. **Cocci** (s., **coccus**) are roughly spherical cells. They can



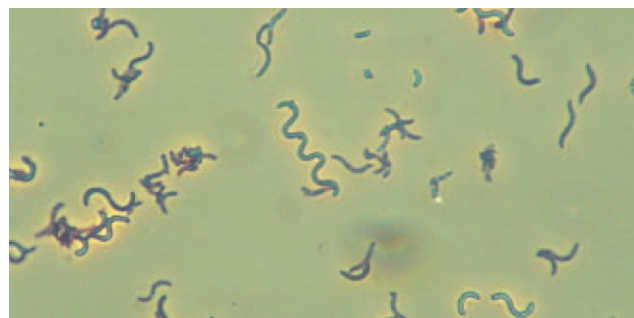
(a)



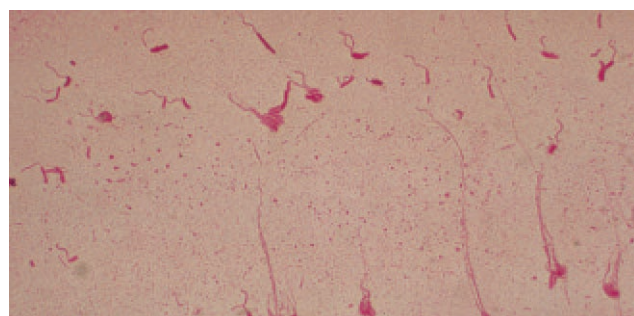
(b)



(c)

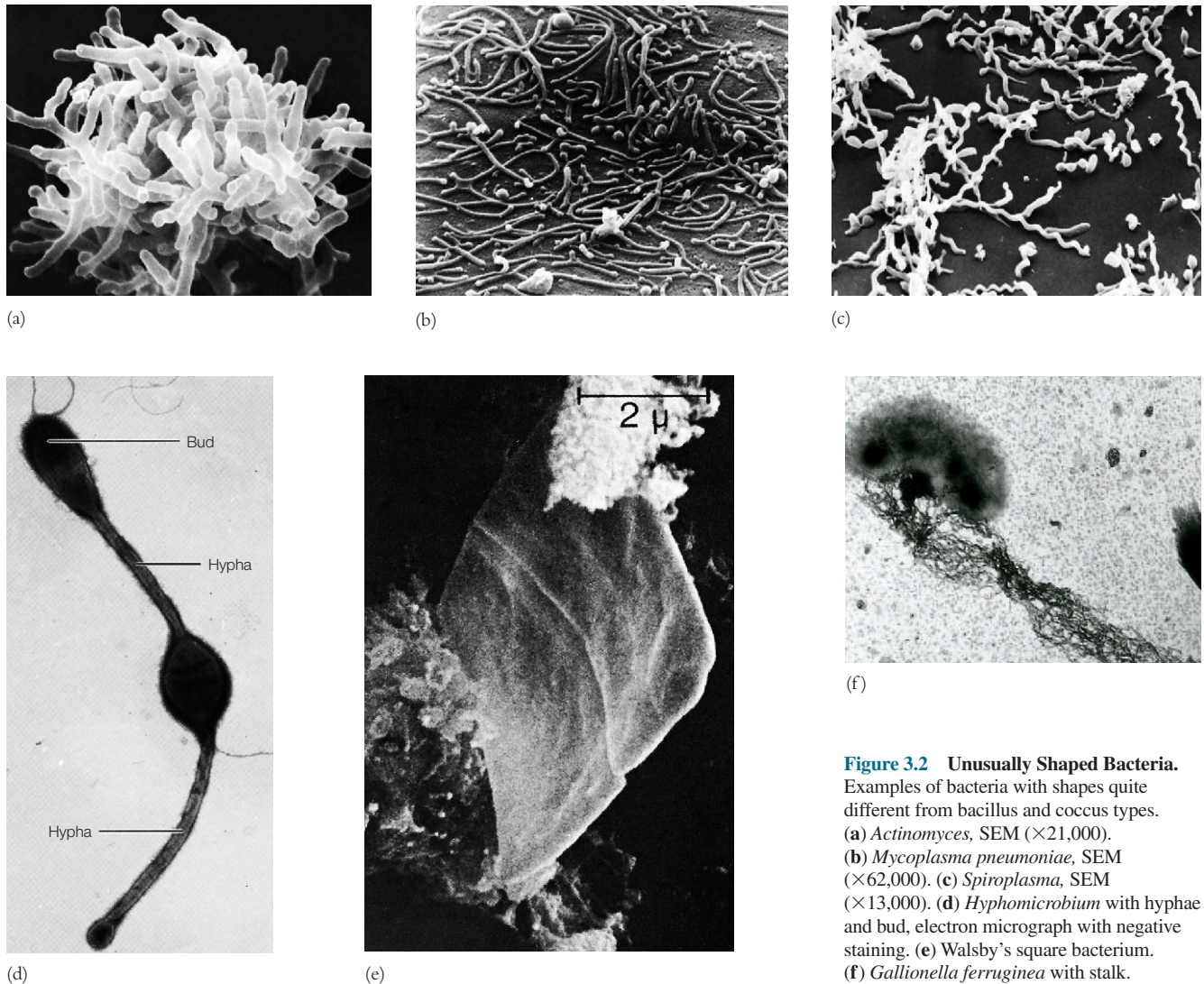


(d)



(e)

Figure 3.1 Representative Bacteria. Stained bacterial cultures as seen in the light microscope. (a) *Staphylococcus aureus*. Note the gram-positive spheres in irregular clusters. Gram stain ($\times 1,000$). (b) *Enterococcus faecalis*. Note the chains of cocci; phase contrast ($\times 200$). (c) *Bacillus megaterium*, a rod-shaped bacterium in chains. Gram stain ($\times 600$). (d) *Rhodospirillum rubrum*. Phase contrast ($\times 500$). (e) *Vibrio cholerae*. Curved rods with polar flagella ($\times 1,000$).

**Figure 3.2** Unusually Shaped Bacteria.

Examples of bacteria with shapes quite different from bacillus and coccus types.

(a) *Actinomyces*, SEM ($\times 21,000$).

(b) *Mycoplasma pneumoniae*, SEM ($\times 62,000$).

(c) *Spiroplasma*, SEM ($\times 13,000$).

(d) *Hyphomicrobium* with hyphae and bud, electron micrograph with negative staining.

(e) Walsby's square bacterium.

(f) *Gallionella ferruginea* with stalk.

exist as individual cells, but also are associated in characteristic arrangements that are frequently useful in bacterial identification. **Diplococci** (s., **diplococcus**) arise when cocci divide and remain together to form pairs (*Neisseria*; see figure 2.15d). Long chains of cocci result when cells adhere after repeated divisions in one plane; this pattern is seen in the genera *Streptococcus*, *Enterococcus*, and *Lactococcus* (figure 3.1b). *Staphylococcus* divides in random planes to generate irregular grapelike clumps (figure 3.1a). Divisions in two or three planes can produce symmetrical clusters of cocci. Members of the genus *Micrococcus* often divide in two planes to form square groups of four cells called tetrads. In the genus *Sarcina*, cocci divide in three planes producing cubical packets of eight cells.

The other common bacterial shape is that of a **rod**, often called a **bacillus** (pl., **bacilli**). *Bacillus megaterium* is a typical

example of a bacterium with a rod shape (figure 3.1c; see also figure 2.15a,c). Bacilli differ considerably in their length-to-width ratio, the coccobacilli being so short and wide that they resemble cocci. The shape of the rod's end often varies between species and may be flat, rounded, cigar-shaped, or bifurcated. Although many rods do occur singly, they may remain together after division to form pairs or chains (e.g., *Bacillus megaterium* is found in long chains). A few rod-shaped bacteria, the **vibrios**, are curved to form distinctive commas or incomplete spirals (figure 3.1e).

Bacteria can assume a great variety of shapes, although they often are simple spheres or rods. Actinomycetes characteristically form long multinucleate filaments or hyphae that may branch to produce a network called a **mycelium** (figure 3.2a). Many bacteria are shaped like long rods twisted into

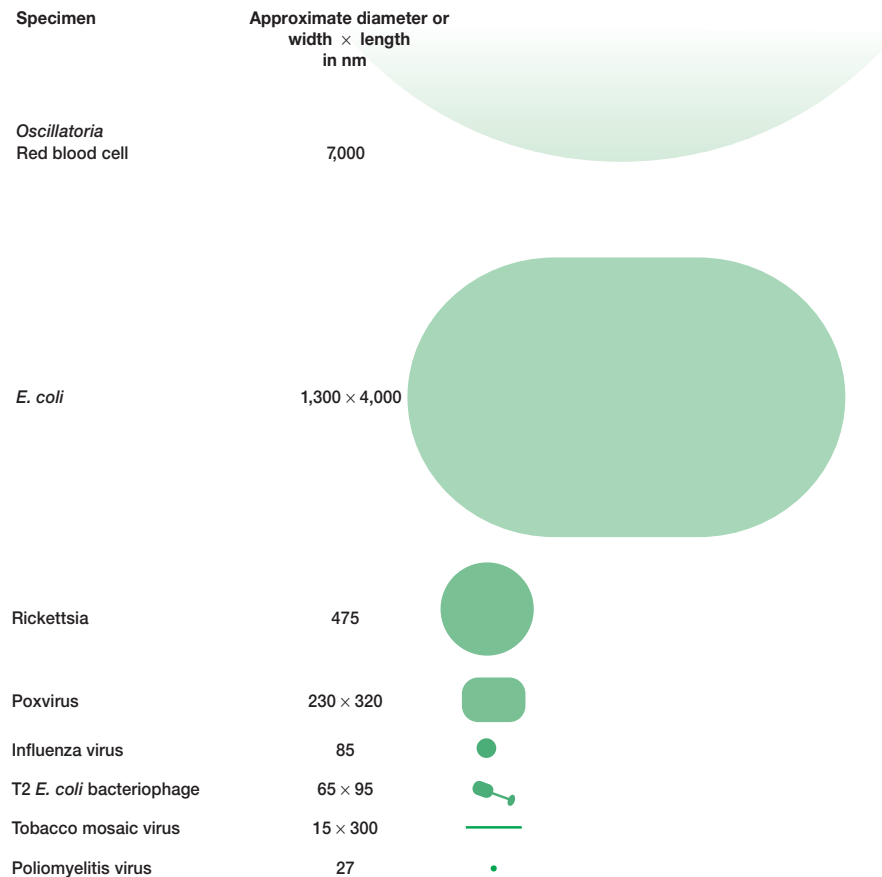


Figure 3.3 Sizes of Bacteria and Viruses. The sizes of selected bacteria relative to the red blood cell and viruses.

spirals or helices; they are called **spirilla** if rigid and **spirochetes** when flexible (figures 3.1d, 3.2c; see also figure 2.8a,c). The oval- to pear-shaped *Hyphomicrobium* (figure 3.2d) produces a bud at the end of a long hypha. Other bacteria such as *Gallionella* produce nonliving stalks (figure 3.2f). A few bacteria actually are flat. For example, Anthony E. Walsby has discovered square bacteria living in salt ponds (figure 3.2e). These bacteria are shaped like flat, square-to-rectangular boxes about 2 μm by 2 to 4 μm , and only 0.25 μm thick. Finally, some bacteria are variable in shape and lack a single, characteristic form (figure 3.2b). These are called **pleomorphic** even though they may, like *Corynebacterium*, have a generally rodlike form.

Bacteria vary in size as much as in shape (figure 3.3). The smallest (e.g., some members of the genus *Mycoplasma*) are about 0.3 μm in diameter, approximately the size of the largest viruses (the poxviruses). Recently there have been reports of even smaller cells. Nanobacteria or ultramicrobacteria appear to range from around 0.2 μm to less than 0.05 μm in diameter.

A few strains have been cultured, but most are simply very small bacteria-like objects only observed microscopically. It has been thought that the smallest possible cell is about 0.14 to 0.2 μm in diameter, but many nanobacteria are reported to be smaller. Some microbiologists think nanobacteria are artifacts, and more research will be required before the significance of these forms becomes clear. *Escherichia coli*, a bacillus of about average size, is 1.1 to 1.5 μm wide by 2.0 to 6.0 μm long. A few bacteria become fairly large; some spirochetes occasionally reach 500 μm in length, and the cyanobacterium *Oscillatoria* is about 7 μm in diameter (the same diameter as a red blood cell). A huge bacterium lives in the intestine of the brown surgeonfish, *Acanthurus nigrofusus*. *Epulopiscium fishelsoni* grows as large as 600 by 80 μm , a little smaller than a printed hyphen. More recently an even larger bacterium, *Thiomargarita namibiensis*, has been discovered in ocean sediment (Box 3.1). Thus a few bacteria are much larger than the average eucaryotic cell (typical plant and animal cells are around 10–50 μm in diameter).

Box 3.1

Monstrous Microbes

Biologists often have distinguished between procaryotes and eucaryotes based in part on cell size. Generally, procaryotic cells are supposed to be smaller than eucaryotic cells. Procaryotes grow extremely rapidly compared to most eucaryotes and lack the complex vesicular transport systems of eucaryotic cells (see chapter 4). It has been assumed that they must be small because of the slowness of nutrient diffusion and the need for a large surface-to-volume ratio. Thus when Fishelson, Montgomery, and Myrberg discovered a large, cigar-shaped microorganism in the intestinal tract of the Red Sea brown surgeonfish, *Acanthurus nigrofusus*, they suggested in their 1985 publication that it was a protist. It seemed too large to be anything else. In 1993 Esther Angert, Kendall Clemens, and Norman Pace used rRNA sequence comparisons (see p. 432) to identify the microorganism, now called *Epulopiscium fishelsoni*, as a procaryote related to the gram-positive genus *Clostridium*.

E. fishelsoni [Latin, *epulum*, a feast or banquet, and *piscium*, fish] can reach a size of 80 μm by 600 μm , and normally ranges from 200 to 500 μm in length (see **Box figure**). It is about a million times larger in volume than *Escherichia coli*. Despite its huge size the organism does have procaryotic cell structure. It is motile and swims at about two body lengths a second (approximately 2.4 cm/min) using the bacterial-type flagella that cover its surface. The cytoplasm contains large nucleoids and many ribosomes, as would be required for such a large cell. *Epulopiscium* appears to overcome the size limits set by diffusion by having an outer layer consisting of a highly convoluted plasma membrane. This increases the cell's surface area and aids in nutrient transport.

It appears that *Epulopiscium* is transmitted between hosts through fecal contamination of the fish's food. The bacterium can be eliminated by starving the surgeonfish for a few days. If juvenile fish that lack the bacterium are placed with infected hosts, they are reinoculated. Interestingly this does not work with uninfected adult surgeonfish.

In 1997, Heidi Schulz discovered an even larger procaryote in the ocean sediment off the coast of Namibia. *Thiomargarita namibiensis* is a spherical bacterium, between 100 and 750 μm in diameter, that often forms chains of cells. It is over 100 times larger in volume than *E. fishelsoni*. A vacuole occupies about 98 percent of the cell and contains fluid rich in nitrate; it is surrounded by a 0.5 to 2.0 μm layer of cytoplasm filled with sulfur granules. The cytoplasmic layer is the same thickness as most bacteria and sufficiently thin for adequate diffusion rates. Nitrate is used as an electron acceptor for sulfur oxidation and energy production.

The discovery of these procaryotes greatly weakens the distinction between procaryotes and eucaryotes based on cell size. They are certainly larger than a normal eucaryotic cell. In addition, some eucaryotic



(a)



(b)

Giant Bacteria. (a) This photograph, taken with pseudo dark-field illumination, shows *Epulopiscium fishelsoni* at the top of the figure dwarfing the paramecia at the bottom ($\times 200$). (b) A chain of *Thiomargarita namibiensis* cells as viewed with the light microscope. Note the external mucous sheath and the internal sulfur globules.

cells have been discovered that are smaller than previously thought possible. The best example is *Nanochlorum eukaryotum*. *Nanochlorum* is only about 1 to 2 μm in diameter, yet is truly eucaryotic and has a nucleus, a chloroplast, and a mitochondrion. Our understanding of the factors limiting procaryotic cell size must be reevaluated. It is no longer safe to assume that large cells are eucaryotic and small cells are procaryotic.

Prokaryotic Cell Organization

A variety of structures is found in procaryotic cells. Their major functions are summarized in **table 3.1**, and **figure 3.4** illustrates many of them. Not all structures are found in every genus. Furthermore, gram-negative and gram-positive cells differ, particularly with respect to their cell walls. Despite these variations procaryotes are consistent in their fundamental structure and most important components.

Procaryotic cells almost always are bounded by a chemically complex cell wall. Inside this wall, and separated from it by a

periplasmic space, lies the plasma membrane. This membrane can be invaginated to form simple internal membranous structures. Since the procaryotic cell does not contain internal membrane-bound organelles, its interior appears morphologically simple. The genetic material is localized in a discrete region, the nucleoid, and is not separated from the surrounding cytoplasm by membranes. Ribosomes and larger masses called inclusion bodies are scattered about in the cytoplasmic matrix. Both gram-positive and gram-negative cells can use flagella for locomotion. In addition, many cells are surrounded by a capsule or slime layer external to the cell wall.

Table 3.1 Functions of Prokaryotic Structures

Plasma membrane	Selectively permeable barrier, mechanical boundary of cell, nutrient and waste transport, location of many metabolic processes (respiration, photosynthesis), detection of environmental cues for chemotaxis
Gas vacuole	Buoyancy for floating in aquatic environments
Ribosomes	Protein synthesis
Inclusion bodies	Storage of carbon, phosphate, and other substances
Nucleoid	Localization of genetic material (DNA)
Periplasmic space	Contains hydrolytic enzymes and binding proteins for nutrient processing and uptake
Cell wall	Gives bacteria shape and protection from lysis in dilute solutions
Capsules and slime layers	Resistance to phagocytosis, adherence to surfaces
Fimbriae and pili	Attachment to surfaces, bacterial mating
Flagella	Movement
Endospore	Survival under harsh environmental conditions

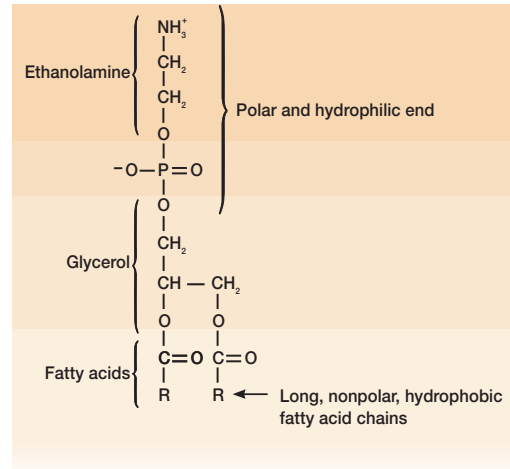


Figure 3.5 The Structure of a Polar Membrane Lipid.

Phosphatidylethanolamine, an amphipathic phospholipid often found in bacterial membranes. The R groups are long, nonpolar fatty acid chains.

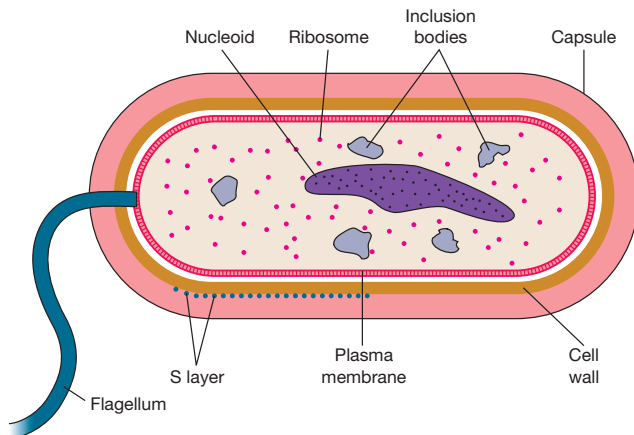


Figure 3.4 Morphology of a Gram-Positive Bacterium. The majority of the structures shown here are found in all gram-positive cells. Only a small stretch of surface proteins in the S-layer has been included to simplify the drawing; when present, these proteins cover the surface.

Prokaryotic cells are morphologically much simpler than eukaryotic cells. These two cell types are compared following the review of eukaryotic cell structure (*see pp. 91–92*).

1. What characteristic shapes can bacteria assume? Describe the ways in which bacterial cells cluster together.
2. Draw a bacterial cell and label all important structures.

3.2 Prokaryotic Cell Membranes

Membranes are an absolute requirement for all living organisms. Cells must interact in a selective fashion with their environment, whether it is the internal environment of a multicellular organism or a less protected and more variable external environment. Cells must not only be able to acquire nutrients and eliminate wastes, but they also have to maintain their interior in a constant, highly organized state in the face of external changes. The **plasma membrane** encompasses the cytoplasm of both prokaryotic and eukaryotic cells. This membrane is the chief point of contact with the cell's environment and thus is responsible for much of its relationship with the outside world. To understand membrane function, it is necessary to become familiar with membrane structure, and particularly with plasma membrane structure.

The Plasma Membrane

Membranes contain both proteins and lipids, although the exact proportions of protein and lipid vary widely. Bacterial plasma membranes usually have a higher proportion of protein than do eukaryotic membranes, presumably because they fulfill so many different functions that are carried out by other organelle membranes in eukaryotes. Most membrane-associated lipids are structurally asymmetric with polar and nonpolar ends (**figure 3.5**) and are called amphipathic. The polar ends interact with water and are **hydrophilic**; the nonpolar **hydrophobic** ends are insoluble in water and tend to associate with one another. This property of lipids enables them to form a bilayer in membranes. The outer surfaces are hydrophilic, whereas hydrophobic ends are buried in the interior away from the surrounding water. Many of these am-

Box 3.2

Bacteria and Fossil Fuels

For many years there has been great interest in the origin of fossil fuels such as coal and petroleum. In the oceans, there is a constant “snow” of prokaryotic membranes and other organic matter that settles on the bottom sediments. Fossil fuel formation begins when organic matter is buried before it can be oxidized to carbon dioxide by microorganisms. When organic matter is buried deeply and subjected to increasing temperature under anaerobic conditions, petroleum and coal are often formed. The quantities involved in these processes are enormous. It has been estimated that the earth contains about 10^{16} tons of carbon in its sediments.

There is increasing evidence that much of the organic material in sediments is bacterial in origin. About 90% of this material is in the

form of insoluble kerogen, an organic precursor of petroleum. Recently the hopanoid bacteriohopanetetrol (figure 3.6b) was isolated from kerogen, and evidence is accumulating that kerogen arises from bacterial activity. We may owe our supply of fossil fuels largely to bacteria that serve as the final degraders of the organic material in dead organisms.

It has been estimated that the total mass of hopanoids in sediments is around 10^{11-12} tons, about as much as the total mass of organic carbon in all living organisms (10^{12} tons). Hopanoids may be the most abundant biomolecules on our planet.

phospholipids are phospholipids (figure 3.5). Bacterial membranes usually differ from eukaryotic membranes in lacking sterols such as cholesterol (figure 3.6a). However, many bacterial membranes do contain pentacyclic sterol-like molecules called hopanoids (figure 3.6b), and huge quantities of hopanoids are present in our ecosystem (Box 3.2). Hopanoids are synthesized from the same precursors as steroids. Like steroids in eukaryotes, they probably stabilize the bacterial membrane. The membrane lipid is organized in two layers, or sheets, of molecules arranged end-to-end (figure 3.7).

Many archaeal membranes differ from other bacterial membranes in having a monolayer with lipid molecules spanning the whole membrane. [Archaea \(chapter 20\)](#)

Cell membranes are very thin structures, about 5 to 10 nm thick, and can only be seen with the electron microscope. The freeze-etching technique has been used to cleave membranes down the center of the lipid bilayer, splitting them in half and exposing the interior. In this way it has been discovered that many membranes, including the plasma membrane, have a complex internal structure. The small globular particles seen in these membranes are thought to be membrane proteins that lie within the membrane lipid bilayer (see figure 2.26). [Freeze-etching technique \(p. 33\)](#)

The most widely accepted current model for membrane structure is the **fluid mosaic model** of S. Jonathan Singer and Garth Nicholson (figure 3.7). They distinguish between two types of membrane proteins. **Peripheral proteins** are loosely connected to the membrane and can be easily removed. They are soluble in aqueous solutions and make up about 20 to 30% of total membrane protein. About 70 to 80% of membrane proteins are **integral proteins**. These are not easily extracted from membranes and are insoluble in aqueous solutions when freed of lipids. [Protein and lipid chemistry \(appendix I\)](#)

Integral proteins, like membrane lipids, are amphipathic; their hydrophobic regions are buried in the lipid while the hydrophilic portions project from the membrane surface (figure 3.7). Some of these proteins even extend all the way through the lipid layer. Integral proteins can diffuse laterally around the surface to new loca-

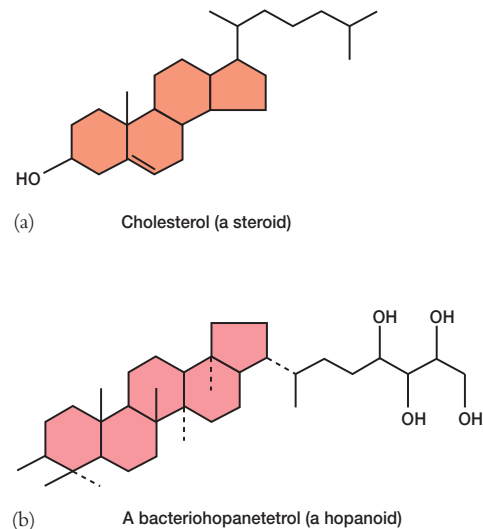


Figure 3.6 Membrane Steroids and Hopanoids. Common examples.

tions, but do not flip-flop or rotate through the lipid layer. Often carbohydrates are attached to the outer surface of plasma membrane proteins and seem to have important functions.

The emerging picture of the cell membrane is one of a highly organized and asymmetric system, which also is flexible and dynamic. Although membranes apparently have a common basic design, there are wide variations in both their structure and functional capacities. The differences are so large and characteristic that membrane chemistry can be used in bacterial identification.

The plasma membranes of prokaryotic cells must fill an incredible variety of roles successfully. Many major plasma membrane functions are noted here even though they are discussed individually at later points in the text. The plasma membrane retains the

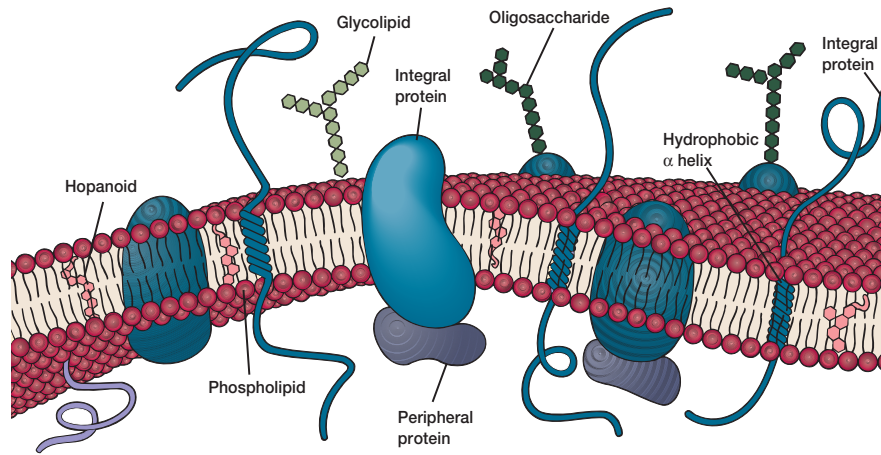


Figure 3.7 Plasma Membrane Structure. This diagram of the fluid mosaic model of bacterial membrane structure shows the integral proteins (blue) floating in a lipid bilayer. Peripheral proteins (purple) are associated loosely with the inner membrane surface. Small spheres represent the hydrophilic ends of membrane phospholipids and wiggly tails, the hydrophobic fatty acid chains. Other membrane lipids such as hopanoids (pink) may be present. For the sake of clarity, phospholipids are shown in proportionately much larger size than in real membranes.

cytoplasm, particularly in cells without cell walls, and separates it from the surroundings. The plasma membrane also serves as a selectively permeable barrier: it allows particular ions and molecules to pass, either into or out of the cell, while preventing the movement of others. Thus the membrane prevents the loss of essential components through leakage while allowing the movement of other molecules. Because many substances cannot cross the plasma membrane without assistance, it must aid such movement when necessary. Transport systems can be used for such tasks as nutrient uptake, waste excretion, and protein secretion. The prokaryotic plasma membrane also is the location of a variety of crucial metabolic processes: respiration, photosynthesis, the synthesis of lipids and cell wall constituents, and probably chromosome segregation. Finally, the membrane contains special receptor molecules that help prokaryotes detect and respond to chemicals in their surroundings. Clearly the plasma membrane is essential to the survival of microorganisms. [Osmosis \(p. 61\); Transport of substances across membranes \(pp. 100–4\)](#)

Internal Membrane Systems

Although prokaryotic cytoplasm does not contain complex membranous organelles like mitochondria or chloroplasts, membranous structures of several kinds can be observed. A commonly observed structure is the mesosome. Mesosomes are invaginations of the plasma membrane in the shape of vesicles, tubules, or lamellae (**figure 3.8** and [figure 3.11](#)). They are seen in both gram-positive and gram-negative bacteria, although they are generally more prominent in the former.

Mesosomes often are found next to septa or cross-walls in dividing bacteria and sometimes seem attached to the bacterial chromosome. Thus they may be involved in cell wall formation

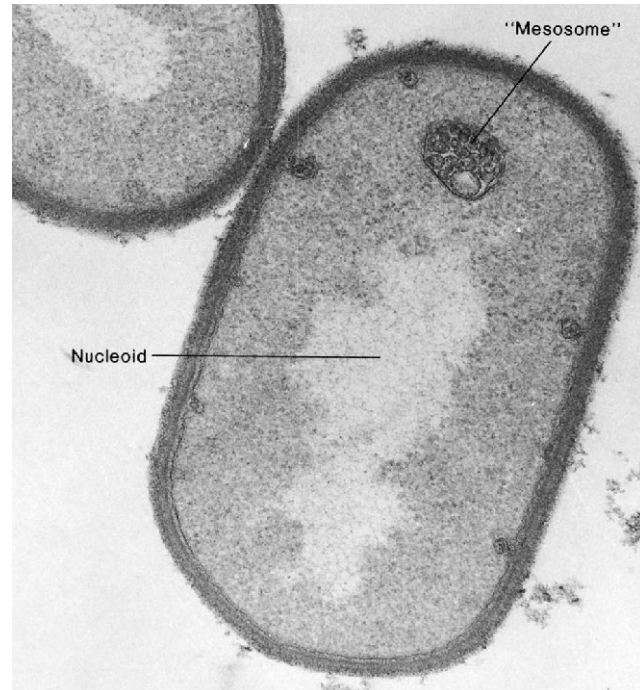
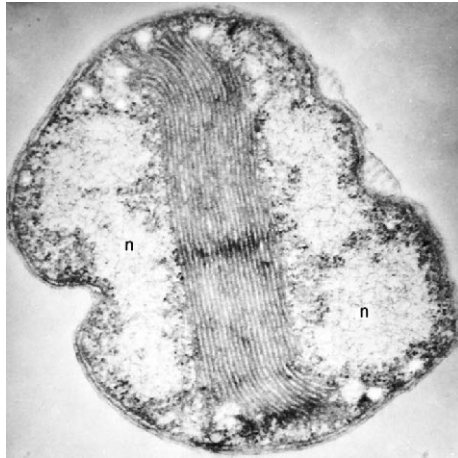


Figure 3.8 Mesosome Structure. *Bacillus fastidiosus* ($\times 91,000$). A large mesosome lies adjacent to the nucleoid.

during division or play a role in chromosome replication and distribution to daughter cells.

Currently many bacteriologists believe that mesosomes are artifacts generated during the chemical fixation of bacteria for elec-



(a)



(b)

Figure 3.9 Internal Bacterial Membranes. Membranes of nitrifying and photosynthetic bacteria. (a) *Nitrocystis oceanus* with parallel membranes traversing the whole cell. Note nucleoplasm (n) with fibrillar structure. (b) *Ectothiorhodospira mobilis* with an extensive intracytoplasmic membrane system ($\times 60,000$).

tron microscopy. Possibly they represent parts of the plasma membrane that are chemically different and more disrupted by fixatives.

Many bacteria have internal membrane systems quite different from the mesosome (figure 3.9). Plasma membrane infoldings can become extensive and complex in photosynthetic bacteria such as the cyanobacteria and purple bacteria or in bacteria with very high respiratory activity like the nitrifying bacteria (see chapter 22). They may be aggregates of spherical vesicles, flattened vesicles, or tubular membranes. Their function may be to provide a larger membrane surface for greater metabolic activity.

1. Describe with a labeled diagram and in words the fluid mosaic model for cell membranes.
2. List the functions of the plasma membrane.
3. Discuss the nature, structure, and possible functions of the mesosome.

3.3 The Cytoplasmic Matrix

Prokaryotic cytoplasm, unlike that of eucaryotes, lacks unit membrane-bound organelles. The **cytoplasmic matrix** is the substance lying between the plasma membrane and the nucleoid (p. 54). The matrix is largely water (about 70% of bacterial mass is water). It is featureless in electron micrographs but often is packed with ribosomes and highly organized (figure 3.10). Specific proteins are positioned at particular sites such as the cell pole and the place where the bacterial cell will divide. Thus although bacteria may lack a true cytoskeleton, they do have a cytoskeleton-like system of proteins in their cytoplasmic matrix. The plasma membrane and everything within is called the **protoplast**; thus the cytoplasmic matrix is a major part of the protoplast.

Inclusion Bodies

A variety of **inclusion bodies**, granules of organic or inorganic material that often are clearly visible in a light microscope, is present in the cytoplasmic matrix. These bodies usually are used for storage (e.g., carbon compounds, inorganic substances, and energy), and also reduce osmotic pressure by tying up molecules in particulate form. Some inclusion bodies are not bounded by a membrane and lie free in the cytoplasm—for example, polyphosphate granules, cyanophycin granules, and some glycogen granules. Other inclusion bodies are enclosed by a membrane about 2.0 to 4.0 nm thick, which is single-layered and not a typical bilayer membrane. Examples of membrane-enclosed inclusion bodies are poly- β -hydroxybutyrate granules, some glycogen and sulfur granules, carboxysomes, and gas vacuoles. Inclusion body membranes vary in composition. Some are protein in nature, whereas others contain lipid. Because inclusion bodies are used for storage, their quantity will vary with the nutritional status of the cell. For example, polyphosphate granules will be depleted in freshwater habitats that are phosphate limited. A brief description of several important inclusion bodies follows.

Organic inclusion bodies usually contain either glycogen or poly- β -hydroxybutyrate. **Glycogen** is a polymer of glucose units composed of long chains formed by $\alpha(1\rightarrow4)$ glycosidic bonds and branching chains connected to them by $\alpha(1\rightarrow6)$ glycosidic bonds (see appendix I). **Poly- β -hydroxybutyrate (PHB)** contains β -hydroxybutyrate molecules joined by ester bonds between the carboxyl and hydroxyl groups of adjacent molecules. Usually only one of these polymers is found in a species, but purple photosynthetic bacteria have both. Poly- β -hydroxybutyrate accumulates in distinct bodies, around 0.2 to 0.7 μm in diameter, that are readily stained with Sudan black for light microscopy and

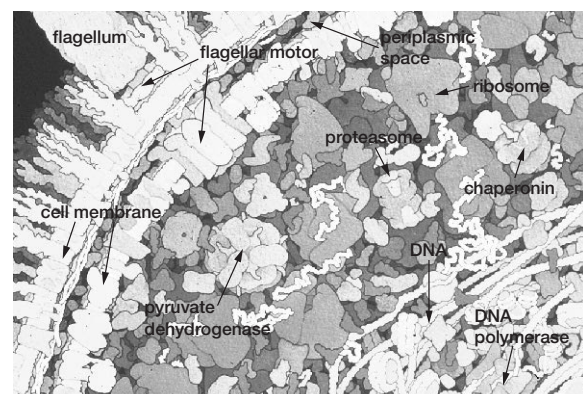
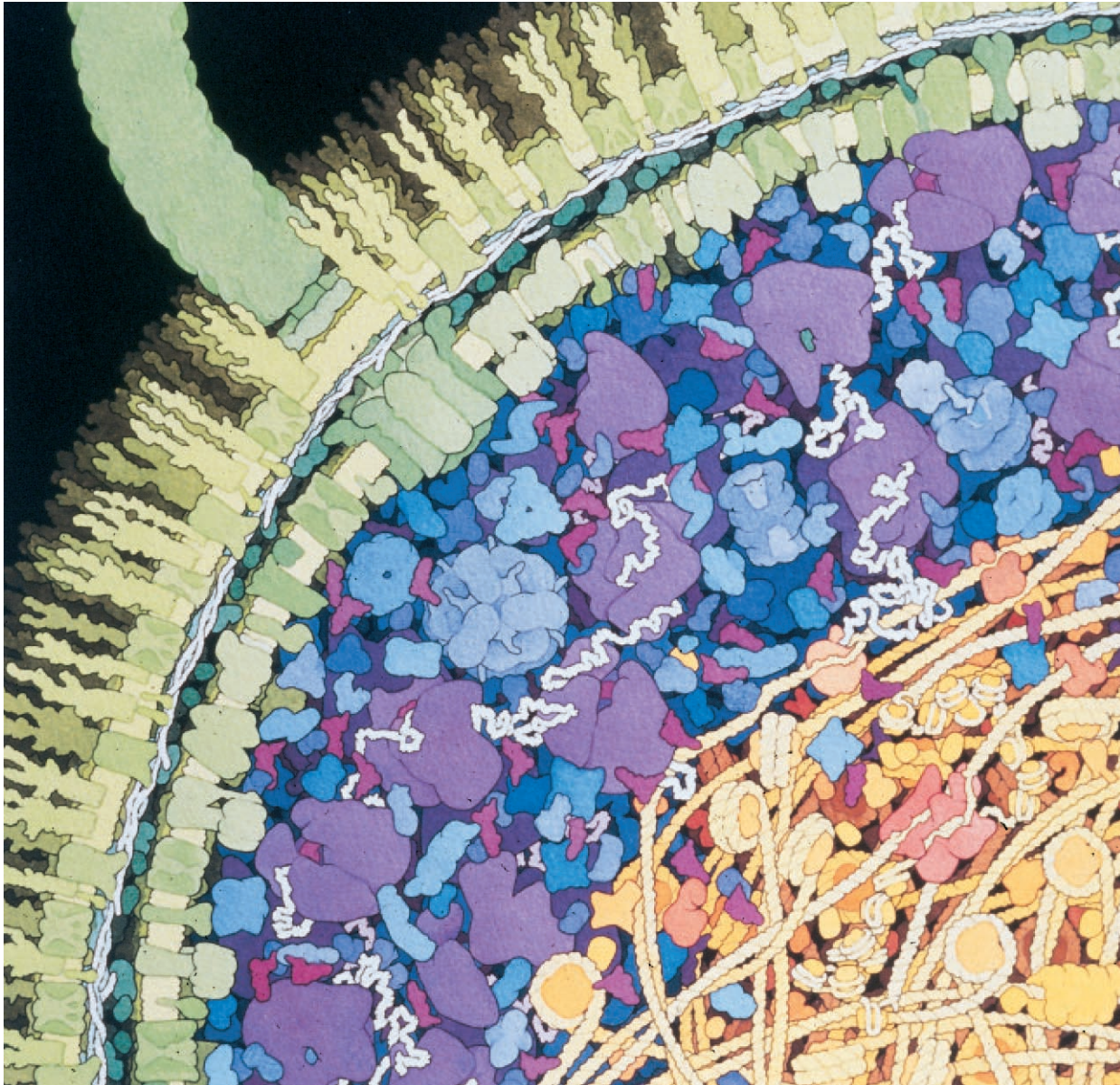


Figure 3.10 A Cross Section of the Bacterium *Escherichia coli* Drawn at a Magnification of a Million Times. The glycocalyx, flagellum, gram-negative cell wall, and plasma membrane are at the top. Ribosomes synthesizing proteins fill the underlying cytoplasmic matrix. At the bottom is the nucleoid with its dense tangle of DNA and associated proteins.

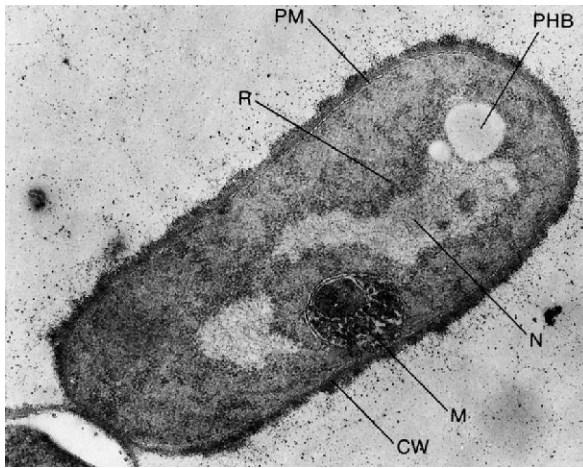


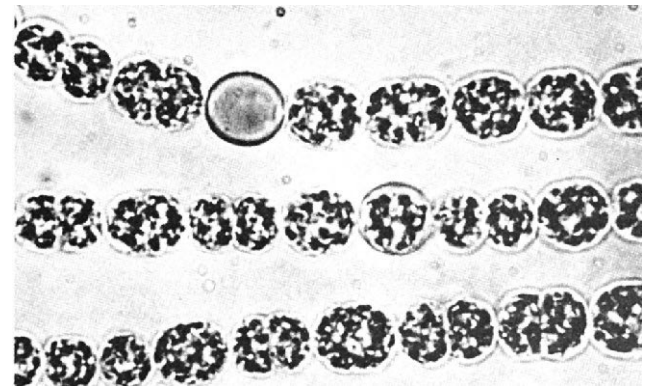
Figure 3.11 The Structure of a Typical Gram-Positive Cell. Electron micrograph of *Bacillus megaterium* ($\times 30,500$). Note the thick cell wall, CW; “mesosome,” M; nucleoid, N; poly- β -hydroxybutyrate inclusion body, PHB; plasma membrane, PM; and ribosomes, R.

are clearly visible in the electron microscope (**figure 3.11**). Glycogen is dispersed more evenly throughout the matrix as small granules (about 20 to 100 nm in diameter) and often can be seen only with the electron microscope. If cells contain a large amount of glycogen, staining with an iodine solution will turn them reddish-brown. Glycogen and PHB inclusion bodies are carbon storage reservoirs providing material for energy and biosynthesis. Many bacteria also store carbon as lipid droplets.

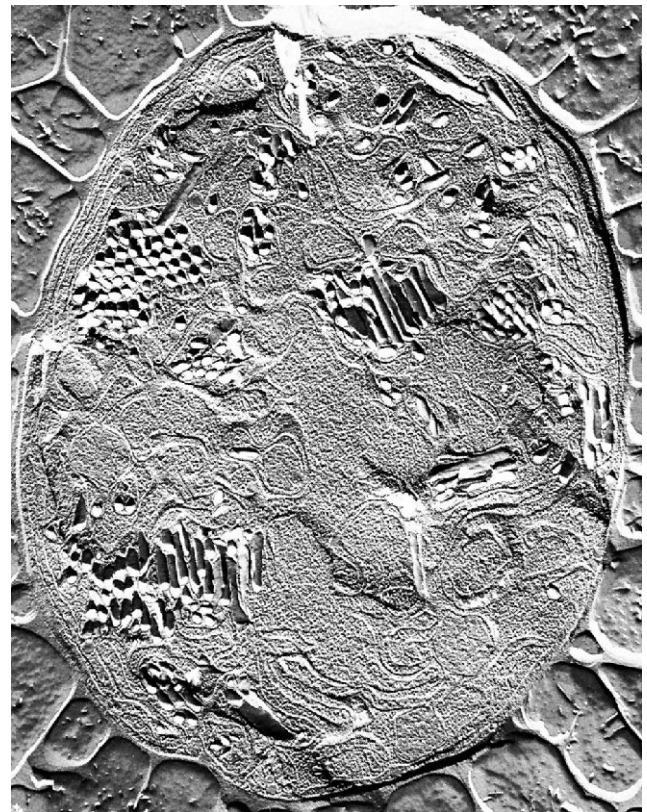
Cyanobacteria have two distinctive organic inclusion bodies. **Cyanophycin granules** (**figure 3.13a**) are composed of large polypeptides containing approximately equal amounts of the amino acids arginine and aspartic acid. The granules often are large enough to be visible in the light microscope and store extra nitrogen for the bacteria. **Carboxysomes** are present in many cyanobacteria, nitrifying bacteria, and thiobacilli. They are polyhedral, about 100 nm in diameter, and contain the enzyme ribulose-1,5-bisphosphate carboxylase (*see p. 208*) in a paracrystalline arrangement. They serve as a reserve of this enzyme and may be a site of CO_2 fixation.

A most remarkable organic inclusion body, the **gas vacuole**, is present in many cyanobacteria (*see section 21.3*), purple and green photosynthetic bacteria, and a few other aquatic forms such as *Halobacterium* and *Thiothrix*. These bacteria float at or near the surface, because gas vacuoles give them buoyancy. This is vividly demonstrated by a simple but dramatic experiment. Cyanobacteria held in a full, tightly stoppered bottle will float, but if the stopper is struck with a hammer, the bacteria sink to the bottom. Examination of the bacteria at the beginning and end of the experiment shows that the sudden pressure increase has collapsed the gas vacuoles and destroyed the microorganisms' buoyancy.

Gas vacuoles are aggregates of enormous numbers of small, hollow, cylindrical structures called **gas vesicles** (**figure 3.12**). Gas vesicle walls do not contain lipid and are composed entirely



(a)



(b)

Figure 3.12 Gas Vesicles and Vacuoles. (a) Filaments of the cyanobacterium *Anabaena flos-aquae* as seen in the light microscope. (b) A freeze-fracture preparation of *Anabaena flos-aquae* ($\times 89,000$). Clusters of the cigar shaped vesicles form gas vacuoles. Both longitudinal and cross-sectional views of gas vesicles can be seen.

of a single small protein. These protein subunits assemble to form a rigid enclosed cylinder that is hollow and impermeable to water but freely permeable to atmospheric gases. Bacteria with gas vacuoles can regulate their buoyancy to float at the depth necessary

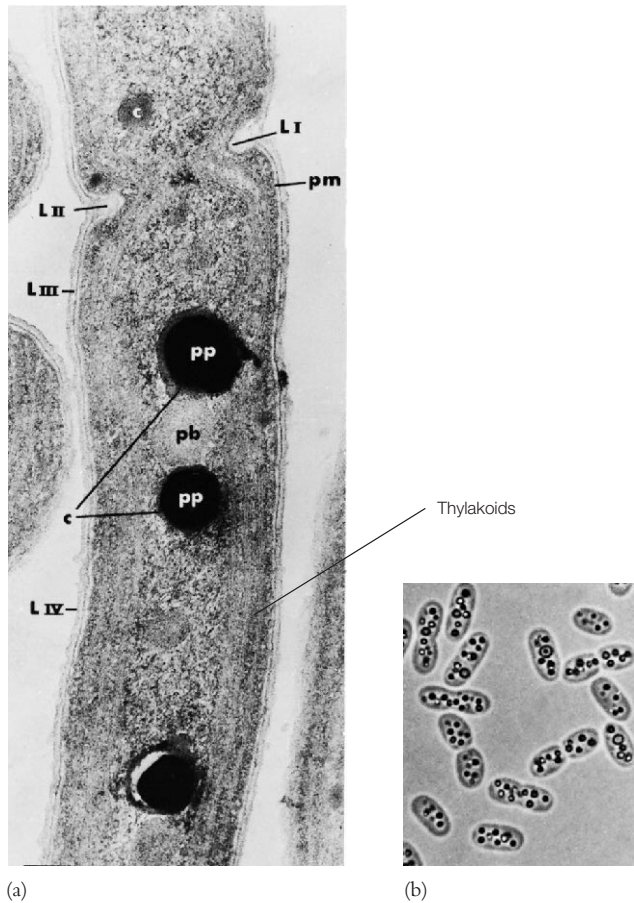


Figure 3.13 Inclusion Bodies in Bacteria. (a) Ultrastructure of the cyanobacterium *Anacystis nidulans*. The bacterium is dividing, and a septum is partially formed, LI and LII. Several structural features can be seen, including cell wall layers, LIII and LIV; the plasma membrane, pm; polyphosphate granules, pp; a polyhedral body, pb; and cyanophycin material, c. Thylakoids run along the length of the cell. Bar = 0.1 μm . (b) *Chromatium vinosum*, a purple sulfur bacterium, with intracellular sulfur granules, light field ($\times 2,000$).

for proper light intensity, oxygen concentration, and nutrient levels. They descend by simply collapsing vesicles and float upward when new ones are constructed.

Two major types of inorganic inclusion bodies are seen. Many bacteria store phosphate as **polyphosphate granules** or **volutin granules** (figure 3.13a). Polyphosphate is a linear polymer of orthophosphates joined by ester bonds. Thus volutin granules function as storage reservoirs for phosphate, an important component of cell constituents such as nucleic acids. In some cells they act as an energy reserve, and polyphosphate can serve as an energy source in reactions. These granules are sometimes called **metachromatic granules** because they show the metachromatic effect; that is, they appear red or a different shade

of blue when stained with the blue dyes methylene blue or toluidine blue. Some bacteria also store sulfur temporarily as sulfur granules, a second type of inorganic inclusion body (figure 3.13b). For example, purple photosynthetic bacteria can use hydrogen sulfide as a photosynthetic electron donor (see section 9.11) and accumulate the resulting sulfur in either the periplasmic space or in special cytoplasmic globules.

Inorganic inclusion bodies can be used for purposes other than storage. An excellent example is the **magnetosome**, which is used by some bacteria to orient in the earth's magnetic field. These inclusion bodies contain iron in the form of magnetite (Box 3.3).

Ribosomes

As mentioned earlier, the cytoplasmic matrix often is packed with **ribosomes**; they also may be loosely attached to the plasma membrane. Ribosomes look like small, featureless particles at low magnification in electron micrographs (figure 3.11) but are actually very complex objects made of both protein and ribonucleic acid (RNA). They are the site of protein synthesis; matrix ribosomes synthesize proteins destined to remain within the cell, whereas the plasma membrane ribosomes make proteins for transport to the outside. The newly formed polypeptide folds into its final shape either as it is synthesized by the ribosome or shortly after completion of protein synthesis. The shape of each protein is determined by its amino acid sequence. Special proteins called molecular chaperones, or chaperones, aid the polypeptide in folding to its proper shape. Protein synthesis, including a detailed treatment of ribosomes and chaperones, is discussed at considerable length in chapter 12.

Note that procaryotic ribosomes are smaller than eucaryotic ribosomes. They commonly are called 70S ribosomes, have dimensions of about 14 to 15 nm by 20 nm, a molecular weight of approximately 2.7 million, and are constructed of a 50S and a 30S subunit. The S in 70S and similar values stands for **Svedberg unit**. This is the unit of the sedimentation coefficient, a measure of the sedimentation velocity in a centrifuge; the faster a particle travels when centrifuged, the greater its Svedberg value or sedimentation coefficient. The sedimentation coefficient is a function of a particle's molecular weight, volume, and shape (see figure 16.7). Heavier and more compact particles normally have larger Svedberg numbers or sediment faster. Ribosomes in the cytoplasmic matrix of eucaryotic cells are 80S ribosomes and about 22 nm in diameter. Despite their overall difference in size, both types of ribosomes are similarly composed of a large and a small subunit.

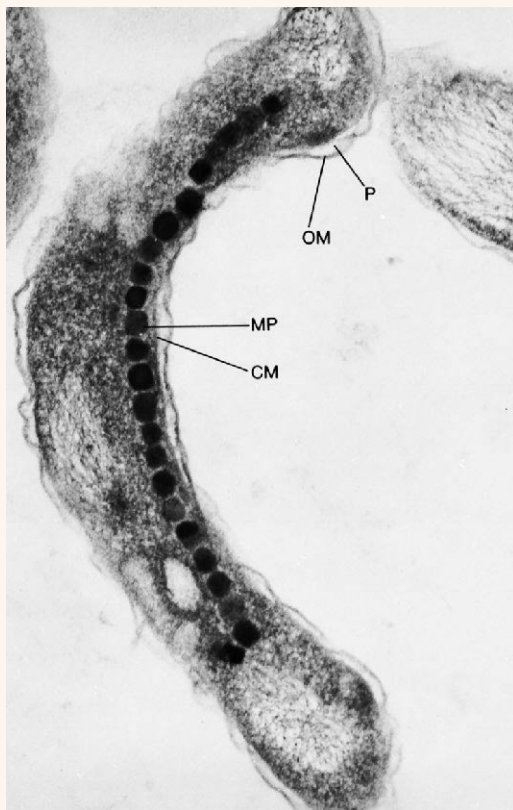
1. Briefly describe the nature and function of the cytoplasmic matrix and the ribosome. What is a protoplast?
2. What kinds of inclusion bodies do procaryotes have? What are their functions?
3. What is a gas vacuole? Relate its structure to its function.

Box 3.3

Living Magnets

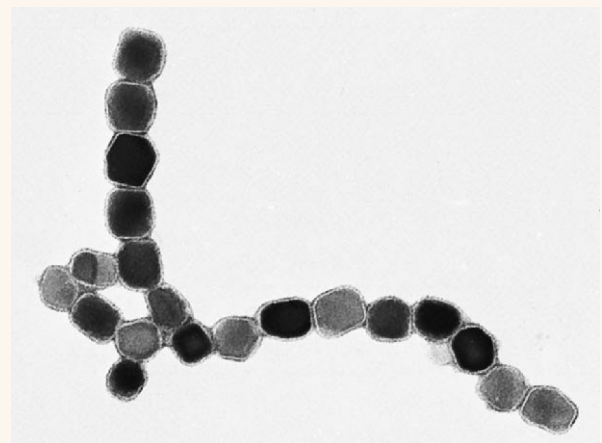
Bacteria can respond to environmental factors other than chemicals. A fascinating example is that of the aquatic magnetotactic bacteria that orient themselves in the earth's magnetic field. Most of these bacteria have intracellular chains of magnetite (Fe_3O_4) particles or magnetosomes, around 40 to 100 nm in diameter and bounded by a membrane (see **Box figure**). Some species from sulfidic habitats have magnetosomes containing greigite (Fe_3S_4) and pyrite (FeS_2). Since each iron particle is a tiny magnet, the Northern Hemisphere bacteria use their

magnetosome chain to determine northward and downward directions, and swim down to nutrient-rich sediments or locate the optimum depth in freshwater and marine habitats. Magnetotactic bacteria in the Southern Hemisphere generally orient southward and downward, with the same result. Magnetosomes also are present in the heads of birds, tuna, dolphins, green turtles, and other animals, presumably to aid navigation. Animals and bacteria share more in common behaviorally than previously imagined.

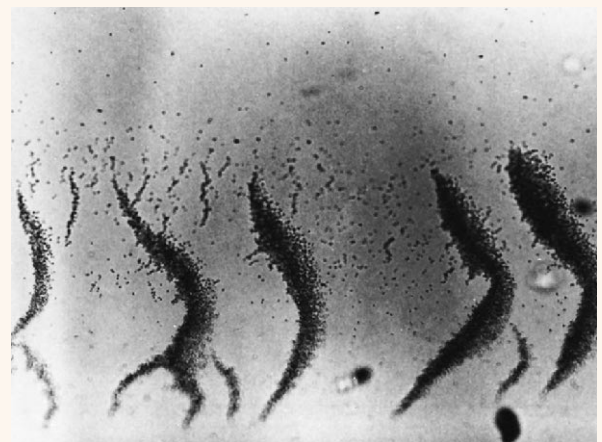


(a)

Magnetotactic Bacteria. (a) Transmission electron micrograph of the magnetotactic bacterium *Aquaspirillum magnetotacticum* ($\times 123,000$). Note the long chain of electron-dense magnetite particles, MP. Other structures: OM, outer membrane, P, periplasmic space; CM, cytoplasmic membrane. (b) Isolated magnetosomes ($\times 140,000$). (c) Bacteria migrating in waves when exposed to a magnetic field.



(b)



(c)

3.4 The Nucleoid

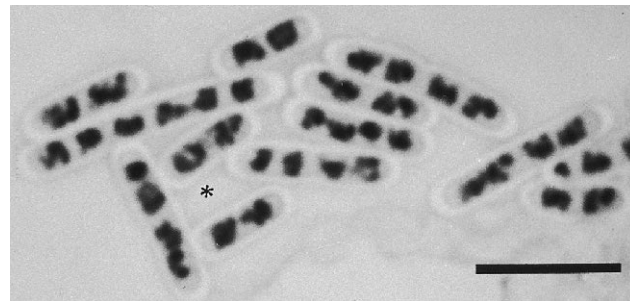
Probably the most striking difference between procaryotes and eucaryotes is the way in which their genetic material is packaged. Eucaryotic cells have two or more chromosomes contained within a membrane-delimited organelle, the nucleus. In contrast, procaryotes lack a membrane-delimited nucleus. The procaryotic chromosome is located in an irregularly shaped region called the **nucleoid** (other names are also used: the nuclear body, chromatin body, nuclear region). Usually procaryotes contain a single circle of double-stranded **deoxyribonucleic acid (DNA)**, but some have a linear DNA chromosome. Recently it has been discovered that some bacteria such as *Vibrio cholerae* have more than one chromosome. Although nucleoid appearance varies with the method of fixation and staining, fibers often are seen in electron micrographs (figure 3.11 and **figure 3.14**) and are probably DNA. The nucleoid also is visible in the light microscope after staining with the Feulgen stain, which specifically reacts with DNA. A cell can have more than one nucleoid when cell division occurs after the genetic material has been duplicated (figure 3.14a). In actively growing bacteria, the nucleoid has projections that extend into the cytoplasmic matrix (figure 3.14b,c). Presumably these projections contain DNA that is being actively transcribed to produce mRNA.

Careful electron microscopic studies often have shown the nucleoid in contact with either the mesosome or the plasma membrane. Membranes also are found attached to isolated nucleoids. Thus there is evidence that bacterial DNA is attached to cell membranes, and membranes may be involved in the separation of DNA into daughter cells during division.

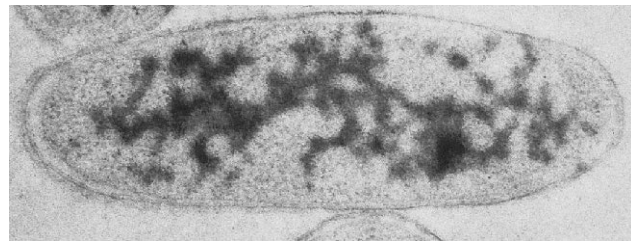
Nucleoids have been isolated intact and free from membranes. Chemical analysis reveals that they are composed of about 60% DNA, 30% RNA, and 10% protein by weight. In *Escherichia coli*, a rod-shaped cell about 2 to 6 μm long, the closed DNA circle measures approximately 1,400 μm . Obviously it must be very efficiently packaged to fit within the nucleoid. The DNA is looped and coiled extensively (see figure 11.8), probably with the aid of RNA and nucleoid proteins (these proteins differ from the histone proteins present in eucaryotic nuclei).

There are a few exceptions to the above picture. Membrane-bound DNA-containing regions are present in two genera of planctomycetes. *Pirellula* has a single membrane that surrounds a region, the pirellosome, which contains a fibrillar nucleoid and ribosome-like particles. The nuclear body of *Gemmata obscuriglobus* is bounded by two membranes (see figure 21.12). More work will be required to determine the functions of these membranes and how widespread this phenomenon is. [The cell cycle and cell division \(pp. 285–86\). Procaryotic DNA and its function \(chapters 11 and 12\)](#)

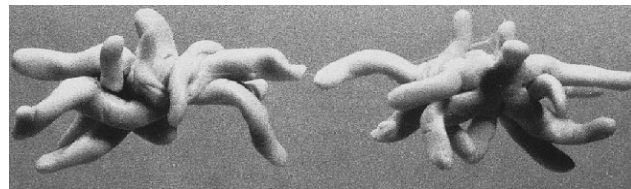
Many bacteria possess **plasmids** in addition to their chromosome. These are double-stranded DNA molecules, usually circular, that can exist and replicate independently of the chromosome or may be integrated with it; in either case they normally are inherited or passed on to the progeny. However, plasmids are not usually attached to the plasma membrane and sometimes are lost



(a)



(b)



(c)

Figure 3.14 The Bacterial Nucleoid. (a) Nucleoids in growing *Bacillus* cells stained using HCl-Giemsa stain and viewed with a light microscope (bar = 5 μm). (b) A section of actively growing *E. coli* immunostained specifically for DNA and examined in the transmission electron microscope. Coupled transcription and translation occur in parts of the nucleoid that extend out into the cytoplasm. (c) A model of two nucleoids in an actively growing *E. coli* cell. Note that a metabolically active nucleoid is not compact and spherical but has projections that extend into the cytoplasmic matrix.

to one of the progeny cells during division. Plasmids are not required for host growth and reproduction, although they may carry genes that give their bacterial host a selective advantage. Plasmid genes can render bacteria drug-resistant, give them new metabolic abilities, make them pathogenic, or endow them with a number of other properties. Because plasmids often move between bacteria, properties such as drug resistance can spread throughout a population. [Plasmids \(pp. 294–97\)](#)

1. Characterize the nucleoid with respect to its structure and function.
2. What is a plasmid?